THE BREAKTHROUGH

Quantifying and Reversing the Bay Area Housing Shortage

This memo seeks to address two questions. First, how much new housing is needed to achieve housing affordability in San Francisco, the Bay Area, and California? Second, where will that new housing be built, and is there even enough developable land available to do so?

The structure of this memo is as follows. First, we define affordability and estimate the amount of housing required to achieve affordability. Second, we quantify how affordability can be achieved through densification or expansion in the Bay Area. Finally, we review some of the major challenges that such a housing construction program would entail.

Quantifying the Shortage

Defining Affordability

We define affordability using the 30% heuristic: households should spend no more than 30% of their pre-tax income on housing. It is debatable as to whether this is the best metric. In particular, many cost-burdened renters choose to live in regions like the Bay Area, accepting high prices in exchange for access to economic opportunity and cultural amenities. However, the 30% rule is a relatively standard threshold for "cost burden."

A region, whether a state, city, or metropolitan statistical area (MSA) is considered "affordable" if the median housing cost is no more than 30% of median pre-tax income. Of course, if a metro area is affordable under this definition, it does not necessarily mean that housing is affordable for all residents.

In assessing affordability, we split the housing market into rentals and owner-occupied housing. For renting to be considered affordable, the median rent should be at most 30% of the median renter's income, and the median cost for owner-occupied housing (mortgage and property tax) should be no more than 30% of the median homeowner income. In measuring median rent, we use the market rate available to a new renter. Due to policies such as rent control, inclusionary zoning, and public subsidies, there might be a large gap between the median rent actually paid and the median market rate rent. We consider the market rate rent to be a more appropriate measure because this is the rate that is available to new renters without waiting lists, lotteries, or other methods of rationing.

Our calculations assume that as new housing is constructed, purchase prices and rents move together. We focus on achieving affordability in the rental market, as a greater share of California renters suffer from cost burden than do homeowners.

We note all instances where limitations in data availability force us to deviate from the above definitions.

Defining the Shortfall and Methods of Estimation

We define several terms that we use in our analysis. Housing is defined in terms of units, and a unit is a single house or apartment. In the United States, the average housing unit is home to 2.58 people. Housing demand in a region is defined to be the number of housing units necessary for the region to be considered affordable using the 30% rule. Housing demand has two components. The present demand is the number of housing units that are required to achieve affordability today, and the annual demand is the number of housing units that need to be built every year to keep real prices constant. We estimate the demand x years in the future to be the sum of the present demand today and x times the annual demand. For example, if a hypothetical region has a present demand in 2017 of 5,000,000 housing units and an annual demand of 50,000 units, then the estimated demand in 2040 is 6,150,000 units.

The housing shortfall is defined as the difference between the number of housing units that exist and the demand. If the hypothetical metro described above actually has 4,500,000 housing units in 2017 and builds 20,000 per year, then the shortfall is 500,000 units in 2017 and will be 1,190,000 units in 2040.

There are several methods of estimating demand and shortfall, and they will be noted below as individual studies are discussed. Our preferred approach uses price elasticity of demand (PED). PED is defined as the percentage change of demand of housing for a 1% change in price. Put another way, if the supply of housing is increased by 1%, then the price will go down by (1/PED)%. If median income, median housing costs, the size of the current housing stock, demand growth, and price elasticity of demand are all known, then demand in the future can be estimated and the present and annual shortfall calculated.

Again, deviations from the above methodology that are required as a result of data availability will be noted.

Data Sources and Results

We draw figures on market rate rentals from a 2017 report by the National Low Income Housing Coalition (NLIHC) and the 2017 Department of Housing and Urban Development (HUD) Fair Market Rents. We use the 2017 HUD figures, rather than the 2018 figures, to be consistent with NLIHC's numbers. The NLIHC's data on income of renting households comes from the Bureau of Labor Statistics.

The target fair market rent in this study is the sum of the following: 45% of the FMR for a two-bedroom apartment, 18% the FMR for a one-bedroom apartment, and 37% the FMR of an efficiency (studio) apartment. These weights are chosen by observing that the household composition of rental households in the United States consists of 37% single people, 18% couples without children, 33% families with children, and 12% other types of households. We assume that single people can get by with studios, couples without children rent one bedroom, and everyone else rents two bedrooms. These assumptions might be a bit stingy, especially for large families that require three or more bedrooms, and so this might cause us to underestimate the housing shortfall slightly.

Data on the current number of housing units is taken from ProximityOne, which in turn calculates based on the US Census and American Community Survey. The Bay Area is defined as the combination of the following metropolitan statistical areas (MSAs): San Francisco-Oakland-Hayward, San Jose-Sunnyvale-Santa Clara, Vallejo-Fairfield, and Santa Rosa. Equivalently, the Bay Area consists of the following nine counties: San Francisco, Alameda, Contra Costa, San Mateo, Marin, Santa Clara, San Benito, Sonoma, and Solano. This definition is consistent with that of the Association of Bay Area Governments. Median wages and median fair market rents for the Bay Area are estimated as a weighted average of wages and FMR for

the MSAs, weighted by population. Note that this is an approximate estimate of the regional median wage and FMR, not an exact calculation.

Region	Monthly Fair Market Rent	Annual Median Renter Household Income	Monthly Affordable Rent	Housing Units
California	$1,339	$42,973	$1,074	13,900,766
Bay Area	$1,938	$60,677	$1,517	2,440,794
San Francisco MSA	$2,110	$57,421	$1,436	1,771,525
San Francisco	$2,501	$78,062	$1,952	386,564

Next, we need an estimate of price elasticity of demand. San Francisco City Economist Ted Egan estimated in 2015 that the price elasticity for market rate housing in San Francisco is 1.35, meaning that a 1% increase in the housing stock should bring prices down by (1/1.35)% = 0.74%. Eric Fisher performs a regression in San Francisco's housing market and finds a price elasticity of demand of 1.7. Jerry Nickelsburg, building from a California Legislative Analyst's Office report, find a price elasticity of demand of 2.0 for California. A report by the California Center for Jobs and the Economy finds a price elasticity of demand of 1.14 for California. We will use a range of elasticities from 1 to 2 and thus report housing shortfall estimates as ranges. We find the following present housing demands and shortfalls.

Region	Current number of units	Present demand (low elasticity)	Present demand (high elasticity)	Shortfall: low elasticity (percent of current units)	Shortfall: low elasticity (percent of current units)
California	13,900,766	17,336,481	21,621,369	3,435,715 (25%)	7,720,603 (56%)
Bay Area	2,440,794	3,118,378	3,984,064	677,583 (28%)	1,543,270 (63%)
San Francisco MSA	1,771,525	2,603,337	3,825,722	831,811 (47%)	2,054,197 (116%)
San Francisco	386,564	495,212	634,396	108,647 (28%)	247,832 (64%)

These figures illustrate very large shortfalls. Even in the low elasticity cases, the housing stocks of San Francisco, the Bay Area, and California would have to increase by at least 25%. In the high elasticity cases, achieving affordability would require at least 50% more housing stock.

Note the apparent paradox that, under both the high and low elasticity scenarios, the shortfall in the San Francisco MSA is greater than that of the full Bay Area, despite the San Francisco MSA being a subset of the Bay Area. This situation occurs because of the higher median wages in the San Jose metro, meaning that if there is enough housing to make the Bay Area affordable as a region, it will not necessarily be affordable to the median renter in the San Francisco MSA.

Our main source for estimating annual demand growth is the California Department of Housing and Community Development (HCD), which bases its forecast on population forecasts from the California State Department of Finance. The forecasts are made with the cohort component model and assume that there will be sufficient housing, infrastructure, and jobs to support the projected population. The forecasts are an annual demand in California of 180,000 units.

The Association of Bay Area Governments constructed a model of employment — estimated through a customized model from Regional Economic Modeling, Inc. — and population growth and estimated that the Bay Area has an annual demand of 27,000 units each year to 2040. The San Francisco-Oakland-Hayward MSA is about 61% of the Bay Area's population, and thus we estimate the annual demand of the MSA to be 16,000, or 61% of 27,000. Similarly, the City/County of San Francisco is about 11% of the population of the Bay Area, and thus San Francisco's annual demand is about 3,100 units per year.

Region	**Present shortfall**	**Annual demand**	**Estimated shortfall in 2040 (percent of current housing stock)**	**New units per year needed each year to 2040 to achieve affordability**
California	3,435,715-7,720,603	180,000	7,575,715-11,860,603 (54-85%)	329,379-515,678
Bay Area	677,583-1,543,270	27,000	1,298,584-2,164,270 (53-89%)	56,460-94,099
SF-Oakland-Hayward MSA	831,811-2,054,197	16,000	1,199,812-2,422,197 (68-137%)	52,166-105,313
San Francisco	108,647-247,832	3100	179,948-319,132 (47-83%)	7,824-13,875

The homebuilding rate required to achieve affordability would be much larger than recent rates. From 2007 to 2016, California added less than 80,000 new housing units per year. Since 2011, the Bay Area has added about 20,000 new housing units per year. San Francisco has added 2,200 new units per year since 2010. These rates are not even enough to match annual demand, let alone to reduce the shortfall.

Additional Estimates of Housing Demand and Shortfall

We highlight some additional notable attempts to estimate the housing shortfall in California and the Bay Area.

One commonly cited source is an October 2016 McKinsey Global Institute report (not commissioned by any external entity). They define a housing gap as the number of new housing units required in California to bring the number of people per unit to the national average — while holding population fixed. The statewide present shortfall is estimated at 2 million units. This estimate is far lower than our estimate of at least 3.4 million. There are two reasons to believe the McKinsey figure is an underestimate. First, it is not clear that McKinsey's metric — people per unit equal to the national average — is adequate, because, as demonstrated in the NLIHC report, the mean renter cannot afford the mean market rate two-bedroom apartment in most states under the 30% rule. Second, if California were to build 2 million new units

quickly, this would likely result in a rise in the state's population and the McKinsey metric would not be achieved.

McKinsey also estimates that an additional 1.5 million new units are needed by 2025 to keep the people per unit ratio the same while accommodating projected population growth, so a total of 3.5 million units are needed by 2025. The implied annual demand is about 170,000 units per year, close to the 180,000 figure reported by HCD, and indeed McKinsey calculated that figure using a similar methodology.

The California Legislative Analyst (LAO) estimated the housing shortage as the number of additional units that should have been constructed between 1980 and 2010 so that the price increases over that period would have been the same as in the rest of the United States. This estimate is made by building an econometric model, based on national data, that estimates housing demand in each county as a function of housing price, housing price in neighboring counties, weather, income, unemployment, education, and other factors. They find that there would be 7 million more people in California today in 2010, or about 20% more. At the current number of people per housing unit, that would be about 2.7 million more units. This model is also likely to underestimate the true housing shortfall. First, the model years are 1980 to 2010, and thus the calculations fail to account for additional shortfall since 2010. Second, in the model's counterfactual scenario, California housing prices in 2010 are 12% less than the actual value, whereas we estimate above that California needs a 20% price drop to achieve affordability.

In 2015, the San Francisco Budget and Legislative Analyst wrote a report on the housing shortfall. Citing the California LAO report mentioned above, they found that to keep the CPI-adjusted rate of price increase in San Francisco at 52% from 1980 to 2010 — which was the overall rate of increase in the United States — rather than the actual increase of 175%, San Francisco should have built 15,300 new units per year. In actuality, the average was 2,011 units per year. In this hypothetical scenario, there would have been 775,608 housing units in San Francisco in 2010 rather than the 376,942 units that there actually existed. This may be an overestimate of the true shortfall, as rents would be 45% less in this scenario, greater than our estimate of the decrease needed to achieve affordability. On the other hand, this estimated shortfall fails to account for annual demand increases since 2010.

Where Will the New Housing Be Built?

Building the estimated demand for housing in the foreseeable future is certainly physically possible. Japan has over three times the population of California and a smaller land area. The city of Tokyo has 85% more people than the Bay Area on less than half the land. The city of Paris has nearly triple the population of San Francisco on slightly less land. While these show that the necessary housing can physically fit in the Bay Area or California, they give us little sense of how or where to build it.

We focus on the Bay Area in reviewing the possibilities for locating new housing. Relative to population, the shortfall in the Bay Area is approximately equal to the statewide shortfall, suggesting that cities all over California face comparable options and challenges.

Bay Area

In the Bay Area, we consider three basic approaches to filling the housing gap: upzoning low density parts of the region, greenfield development, and development around transit.

The nine-county Bay Area has 1,580 census tracts, as given by the Metropolitan Transportation Commission. Of these, the median density is 11.3 people per acre, about the density typical of a

neighborhood of detached single family homes (choose settings Western / Low / Suburban), and 49% of the region's population lives in tracts with lower density. If these lower density tracts were to increase their housing stocks by 106-178%, it would be enough to close the estimated shortfall in 2040. Since the MTC reports population and not housing units per census tracts, these calculations make the simplifying assumption that the average number of people per unit is the same between low density and high density tracts.

As for greenfield development, the Greenbelt Alliance has identified 63,500 acres of land in the Bay Area at high risk of development in the next 10 years, 293,100 acres at risk over the next 30 years, and a total of 2,439,100 acres that could be developed further into the future. While it is not clear how likely any of development actually is, we use these figures for illustrative purposes. If all of the present and annual housing shortfall by 2040 is met on 63,500 acres, this land would have to be developed to a density of 20-34 units per acre, characteristic of low-rise apartments (choose settings Western / High / Urban). If the shortfall is to be met on 293,100 acres, the land would have to be developed to 4.4-7.4 units per acre, characteristic of dense suburbs of detached houses (choose settings Western / Low / Urban). If the shortfall is to be met on the full 2,439,100 acres available, it would have to be developed to an average density of 0.5-0.9 units per acre, characteristic of low-density suburbia.

Acres of greenfield development	**Density required to meet projected shortfall by 2040, units per acre**	**Neighborhood character**
63,500	20-34	Low-rise apartments
293,100	4.4-7.4	Suburbs
2,439,100	0.5-0.9	Sparse suburbs

The figures above contrast to 1,192,900 acres of permanently protected land in the Bay Area that are not considered developable, 790,300 acres that are already urbanized, and 95,408 acres by which the urbanized portion of the Bay Area expanded from 1990 to 2014.

Wendell Cox estimates that there are 1,500-4,300 square miles (0.96 to 2.75 million acres) of developable land in the Bay Area, the upper bound of which is close to the Greenbelt Alliance's estimate.

A third approach is upzoning around transit. The McKinsey Global Institute report estimates that there are about 250 transit-rich square miles in California, or about 80 square miles in the Bay Area. This is 6% of the urbanized portion of the Bay Area. A transit-rich area is developable land within a half mile of a high-frequency transit hub. To fill the total housing shortfall in 2040, it would be necessary to increase the density in these transit-rich areas by 25-42 units per acre, roughly the equivalent of *adding* new low- to mid-rise apartments on top of what is already there. This would be 27-112% more than what MGI estimates as the potential for these zones, which is approximately doubling the density. The MGI transit-rich upzoning proposal is similar to the potential effect of SB 827, a map of which can be seen here. Almost all of San Francisco and a large fraction of Oakland and Berkeley would be upzoned to allow buildings between 5 and 8 floors. Upzoning means that cities cannot legally prevent buildings from being constructed, but it does not necessarily mean that they will be built. The actual number of new units built will probably be much less than the zoned capacity because the rate of development is limited by ownership turnover, economic considerations, and other factors.

San Francisco-Oakland-Hayward MSA

If the San Francisco-Oakland-Hayward MSA is taken in isolation, the challenge of filling the housing shortfall becomes much greater with a higher need and less area to work with.

Half of the population of the MSA lives in census tracts with a population density less than 11.2 people per acre. It would be necessary to increase the density of these tracts by 94-232% by 2040 to achieve affordability.

Alternately, the Greenbelt Alliance identifies the following developable greenfield land in the five county MSA.

County	**Developable land, acres (high risk)**	**Developable land, acres (medium risk)**	**Developable land, acres (low risk)**
Alameda	6,300	23,300	183,100
Contra Costa	20,000	42,000	113,300
Marin	2,400	9,200	87,600
San Mateo	4,700	18,000	78,800
San Francisco	0	0	0
Total	33,400	92,500	462,800

Meeting all the estimated shortfall in 2040 through greenfield development would require development to the following densities, depending on the number of acres converted.

Acres of greenfield development	**Density required to meet projected shortfall by 2040, units per acre**	**Neighborhood character**
33,400	36-73	Mid-rise apartments
92,500	10-19	Low-rise apartments
462,800	2-4	Detached single-family

San Francisco

There is no significant greenfield development potential in San Francisco, so achieving affordability through development within the city necessarily requires increasing density.

The median population density of census tracts in San Francisco is 40 people per acre, typical of low-rise apartments. Increasing the density of all tracts with fewer than 40 people per acre by 56-128% by 2040 would close the housing shortfall.

Redevelopment of current multifamily housing to the maximum allowed by current zoning, along with a strategic upzoning, could also help meet the shortfall. The McKinsey Global Institute report estimates that in San Francisco, an additional 70,000 units would be possible by using existing multifamily-zoned parcels to capacity. In San Francisco, 31% of land zoned for multifamily housing is built at less than 50% capacity, and those built above 50% of capacity are not considered economic to redevelop. A revision of the Central Soma Plan could add an additional 33,000 new units. Combined, these two actions would add a third to a half of the projected shortfall by 2040.

Additional Estimates of Potential Housing Supply

The McKinsey Global Institute report identifies tools to build 2.9-5.6 million new units in California. These include 1.2-3.0 million new units that could come from intensifying building around transit, 225,000 on vacant land zoned for multifamily residential, 800,000 by adding units to existing single family homes (e.g. accessory dwelling units, or "granny flats"), 1 million by adding to existing multifamily-zoned land, and 600,000 by extending single family homes into greenfield development that is adjacent to current urban land. Even the high number, 5.6 million new units, is not sufficient to satisfy our estimated shortfall in 2040 under any scenario.

The new housing around transit stations is based on an analysis of 1,083 high-frequency transit stations and 12 high speed rail stations. The 12 HSR stations are supposed to support 40,000 new units total. The upper end of the given range — 3 million units — would increase total population near transit by a factor of 3.6 and upzone these areas to anywhere from 10 to 70 units per acre, depending on existing characteristics of the site. Ten units per acre is a mix of single-family homes and townhomes, while 70 units per acre indicate mid-rise apartments, as pictured here.

The San Francisco Housing Pipeline is a useful tool to track new development in the city. As of December 2017, there are 63,353 new net housing units in the pipeline, which includes units that are under construction, that have been approved for construction, and that have not yet been approved but for which the application has been filed. Based on past history, many of these units will not actually be built. Very large projects — Candlestick Park, Treasure Island, and Parkmerced for instance — generally have timelines over a decade. It is therefore safe to assume that there will not be a dramatic acceleration of new housing in the next 10 years, even if zoning and planning is liberalized tomorrow.

In 2016, the Bay Area Council Economic Institute modeled 10 policies that were under serious discussion at the state or local level at the time. Their cumulative effect would be to make San Francisco affordable to 77,335 more households than would be the case absent the policies (this figure was calculated by summing the effect of each policy; this might overestimate the true figure by double-counting). They also modeled the impact of 10 harmful policies, one of which, 25% inclusionary zoning, has since been adopted and appears to be having negative effects.

Tradeoffs and Challenges to Growth

Achieving housing affordability in the Bay Area, though not impossible, is a major, multi-decade challenge that will necessarily entail major tradeoffs.

Objections to New Development

Doubling the density of half the census tracts in San Francisco, the MSA, or the full Bay Area would be a major transformation of the region. This does not necessarily mean that every single family home is

replaced by a duplex, but it does likely mean that there would be new duplexes or apartments in almost every block, and nearly every resident of the region would see development within a short walking distance. Alternately, it is possible to reduce the number of residents affected by concentrating development in particular parts of the city, but this leads to objections that certain neighborhoods are "dumped on" and that the process lacks fairness.

Common objections to new infill development include shadows, difficulty in finding parking, and changes to neighborhood character. Advocates of development often regard such objections as frivolous or selfish, in light of the housing shortage. However, the interests of existing residents, who have a deep economic and personal stake in the city, should not be ignored.

Greenfield development does not necessarily avoid land use conflicts. There are legitimate public interests in urban containment, such as preservation of green space and local farming. Greenfield development also raises environmental concerns, such as local biodiversity loss and greenhouse gas emissions from long-distance transportation. These concerns should also not be ignored.

Transportation Challenges

No matter where new development occurs, growth places a strain on transportation networks. Unless transportation infrastructure is improved, new development will result in more traffic congestion and crowding in public transit. Residents in new greenfield development will likely have longer daily commutes.

From 1980 to 2015, average door-to-door commute time in the Bay Area, by all modes, has increased from 24 to 31 minutes. Average transit commute time increased from 45 to 51 minutes from 2006 to 2015. The Bay Area is now close to the median average commute times among major US metros, and in turn the national trend has been toward longer commute times since 1980. According to the INRIX 2017 Global Traffic Scorecard, San Francisco has the 5th worst traffic congestion among 1,360 world cities studied, measured by average number of hours (79) that commuters spend in traffic jams each year.

BART (Bay Area Rapid Transit) is also struggling to keep up with demand. As of 2015, crowding during rush hour commuting was 20% above federal recommendations. Caltrain is also struggling with crowding. The theory behind boosting the housing supply around transit, as envisioned by SB 827, is that transit-oriented development can support a higher population density than automobile-dependent neighborhoods. However, it is not clear that either BART or Caltrain can handle a major increase in demand.

With the transportation system struggling to support even the development that has occurred since 1980, it appears that either major infrastructure improvements or new transportation technologies are necessary to support the accelerated development required to achieve housing affordability.

Economic Challenges and Considerations

There are economic challenges associated with rapid new housing development. In 2017, the professional services company Turner & Townsend identified San Francisco as the second most expensive market for construction among major cities in the world (behind New York). Taller buildings generally cost more per square foot. On the other hand, sprawl and greenfield development tend to have higher per capita infrastructure costs. Over the next 20 years, there are likely to be periods of recession or high interest rates that will slow construction. Thus it is unclear if the high rates of new housing construction needed to achieve affordability are economically feasible, even absent zoning and planning restrictions.

Since rapid housing development must necessarily be accompanied by rapid infrastructure development, temporary tax increases are probably needed.

Housing markets are tightly coupled, with people commuting between cities around the Bay Area. While housing gap estimates are a useful way to think of each city's contribution to the problem and solution, no Bay Area city, even San Francisco, can unilaterally control its own market. Furthermore, since companies and people can move around the country, there is a national component to the housing market. With global capital flows, there may even be a global component to the housing market.

Our analysis has assumed that as new housing construction occurs, housing prices go down and salaries are unaffected. In reality, the effect is unclear. Agglomeration economies are a fundamental concept in urban economics. The reason people want to live in the Bay Area in the first place, despite the high price, is access to the industry, culture, and amenities that a wealthy and dense population can provide. Would building more housing in the Bay Area increase economic and cultural opportunity, thereby increasing the desirability of the region and negating some of the expected price reduction from an increased housing supply (an urban analogue of the rebound effect)? Some skeptics of new housing make a variant of this argument: that building more housing could created "induced demand", attracting new people whose demand for services drives up the price rather than lowers it. There are positive aspects of agglomeration economies: increased wealth in the Bay Area and a reduction of pressure on the housing markets in other parts of the country (though other cities could then experience an agglomeration diseconomy).

Alternately, it is possible that rapid housing construction will result in an influx of new workers, driving down salaries as they compete for jobs. Similarly, if housing costs come down, this may attract a new wave of lower income workers who in turn become cost-burdened. Thus affordability is a moving target.

It must also be acknowledged that future housing demand projections are highly uncertain. Future demand will be influenced by factors beyond the control of housing policy, such as recessions, the health of the tech industry, or development in the rest of the country.

Another major confounding factor in this analysis is that housing hardly functions as a free market in California. About 75% of San Francisco's rental stock is rent controlled, with additional units subsidized in some way, and other major cities in California have rent control policies. Proposition 13 also distorts the housing market by creating a disincentive for homeowners to sell. It is not clear what impact these policies have on the observed affordability gap, nor is it fully clear what effect modifying or repealing them would have on affordability.

Preliminary Conclusions

The San Francisco Bay Area and other fast-growing regions in the state face daunting, if not intractable, housing shortages. Given present trends, these regions appear unlikely to keep up with continuing housing demand growth, much less build enough to make a significant dent in the affordability crisis. Considered separately, neither traditional affordable housing remedies, nor the proposals of the YIMBY movement, nor those of urbanists such as Joel Kotkin and Wendell Cox appear likely to offer relief at an order of magnitude that would be necessary to rectify the state's housing crisis.

Absent draconian measures (e.g. eminent domain or similar measures to dramatically transform existing single family low density areas into much higher density development) it will likely prove difficult to increase density within the existing urban footprint at a pace consistent with reducing housing costs in many impacted regions. Without paving over much of the green space in and around existing

metropolitan regions, expanding low density single family development into adjacent greenfields cannot produce sufficient new housing to significantly reduce regional housing costs. And while the state does not lack for space where additional housing might be build further afield, making those areas accessible to the employment centers that have made housing scarce, regions that are magnets for in-migration would require dramatic expansion of costly transportation infrastructure. This expansion would include high speed transit connections, which the state has proven unable to build at acceptable cost.

As with so much else, California's housing crisis has become a screen upon which all sorts of competing values and agendas have been projected. For urbanists, the solution is to build more infill development. For suburbanists, the solution is more single family greenfield development. For developers, it is to reduce zoning and other regulatory barriers. For affordable housing advocates, it is more public funding to subsidize below market rate housing.

Tackling the state's housing crisis will likely require giving all parties much of what they want, while also demanding concessions from all as well — a substantial increase or reform in public support for below market rate housing, paid for through mechanisms upon which the construction of new market rate housing is not predicated; upzoning and sweeping reform of CEQA to expedite the permitting and construction of new housing, both infill and greenfield; opening up new greenfields close to urban centers to dense multi-unit development; and very significant expansion of new high-speed transportation infrastructure, whether rail or something else, to bring more developable land within commute distance of urban employment centers.

All of the analysis and conclusions included in this memo are preliminary and most require further analysis, to better understand the underlying data and trends upon which the analysis is based. We have not looked carefully into how demand growth projections have been made, how demand elasticities are calculated, or how developable land both within and beyond various urban boundaries have been calculated. As noted at the beginning, these initial calculations and analyses are intended to provide an initial order of magnitude sense of how substantial the challenge is and what sorts of levers are likely to be large enough to meaningfully affect it.

In a forthcoming memo, we will take a careful look at the political environment, how major interests in the state are arrayed around the key levers that might make a difference, what the possibilities are for new strategies, framings, and coalitions that might disrupt the current stasis, and what the prospects are for funding continuing work and building productive alliances with key players in the space.